

Per-backend telemetry in
latency_ms, queue_depth,
health flag
(sorted by backend_id,
first N_MAX slots)

any eligible
backend?
(healthy or
degraded)

no

yes

Empty shadow action
(no rankings emitted)

Build eligibility mask:
healthy / degraded kept,
unhealthy / unknown
excluded

Update running-min latency
per eligible backend
(capacity estimate,
floored at cap_floor_ms)

base = running-min
(past latencies only),
cap = 1 / base

Score each backend
score = cap /
(lat/base)^degr_pow
(monotone transform:
falls as current lat rises)

lat >
cut x min
eligible lat?

no

yes

Demote far-outlier:
score x 1e-3

Normalise masked scores
into target weights
(excluded set to 0,
sum = 1)

total queue_depth
< idle_load?

yes

no

a = 1.0
(snap to target)

a = alpha
(damped update)

EMA weight update
 $w = (1 - a) * w + a * \text{target}$,
renormalise over mask

Emit routing weight vector
rankings for eligible $w > 0$

operating_mode
= hybrid?

no (default)

yes

Publish in shadow
(observe-only,
smartload.routing)

Publish active
(sidecar may apply)